

Background

Wound regeneration is a critical survival process that allows organisms to repair damaged tissue after injury.

The model organism *Drosophila melanogaster* (fruit fly) is widely used in regeneration research because many of its cellular and genetic mechanisms are highly conserved with those of humans. In addition, its short life cycle, well-mapped genome, and advanced genetic tools make it ideal for studying how specific genes influence biological processes such as wound healing.

During epithelial wound repair in *Drosophila* larvae, cells surrounding the injury rapidly reorganize actin filaments to migrate toward the wound site and close the damaged area. This actin remodeling is regulated by Rac1, a small GTPase that controls actin polymerization, branching, and cytoskeletal organization.

To investigate the effects of Rac1 on regeneration, the UAS/GAL4 system can be used to selectively overexpress Rac1 in larval epidermal tissue. This genetic system enables researchers to activate specific genes in targeted cells, allowing direct analysis of how altered Rac1 activity impacts wound healing rates and epithelial repair.



The significance of this project is that wound repair is essential for survival, many of the cellular processes that regulate efficient healing are still not fully understood. This project is also significant because the biological pathways involved in wound healing are highly conserved across species, meaning discoveries in *Drosophila* can provide insight into similar processes in humans. Additionally, impaired wound healing is associated with many medical conditions, including chronic wounds, diabetes-related complications, and tissue damage after injury or surgery.

References

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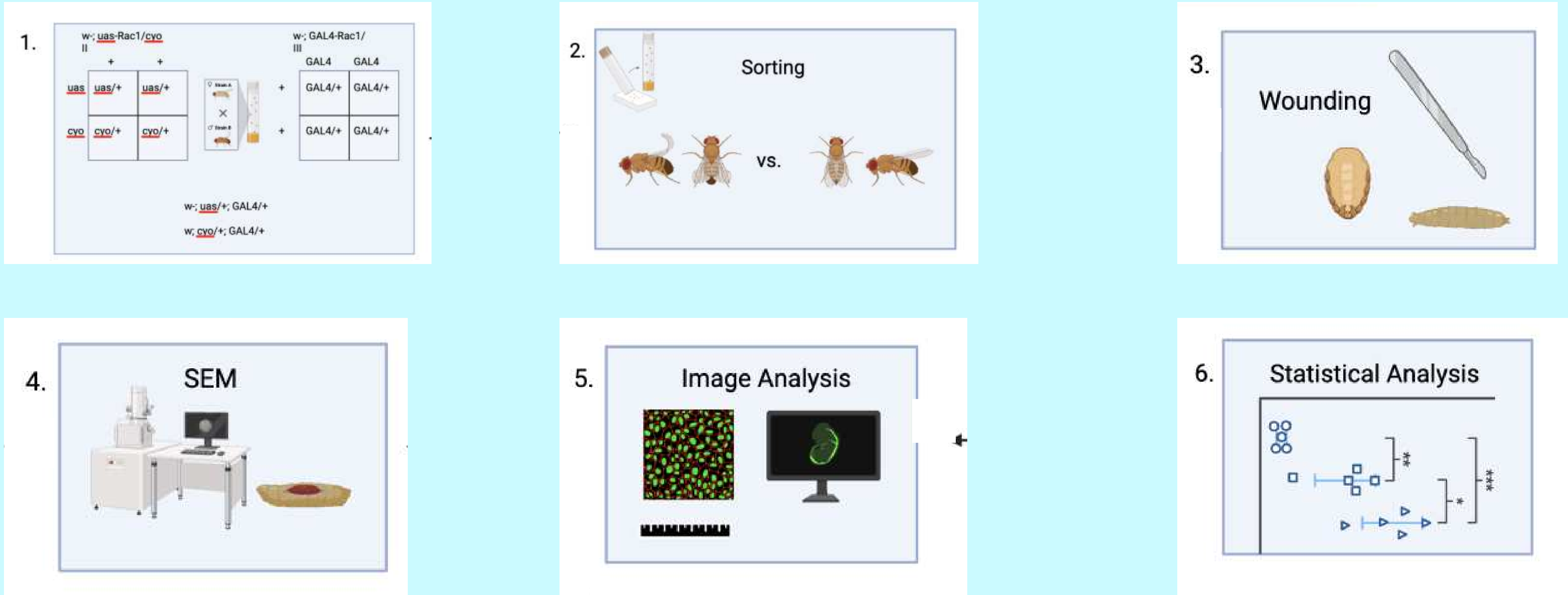
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Rebuild, Renew, Recover: Testing the Overexpression of RAC1 in Wound Regeneration of Drosophila Melanogaster

Research Question: Does overexpression of Rac1 affect wound regeneration factors, such as time and width, and polymerization during the wound regeneration process in a *Drosophila melanogaster* model?

Hypothesis: If I overexpress the RAC1 gene in *Drosophila Melanogaster*, then their regeneration process will show faster healing in the actin dynamics of the fly because RAC1 drives cell behaviors needed for repair.

Experimental Design



1. The first step will be to cross the flies. This will be done by putting 5-6 female virgin flies into a vial with 4 male flies that do not have to be virgin.
2. After the crossing I need to do CO2 sorting to identify the undesired flies, which will be the ones with curly wings, and discard them because they cannot be used for experimentation.
3. Wait for hatched eggs to become larvae, which is from 1-4 days and that is prime wounding days. Cold sort flies and use fine forceps to orient the larva and puncture the larval abdomen at a consistent ~20–30 μm deep for each fly.
4. Transfer the wounded larvae back to a food vial for observation.
5. Peel off any scabbing on the wound. Take several pictures, especially of the wounds in separate time intervals using a microscope camera
6. Analyze images taken by the microscope using image analysis software, specifically ImageJ.
7. Conduct a statistical Analysis using an independent t-test

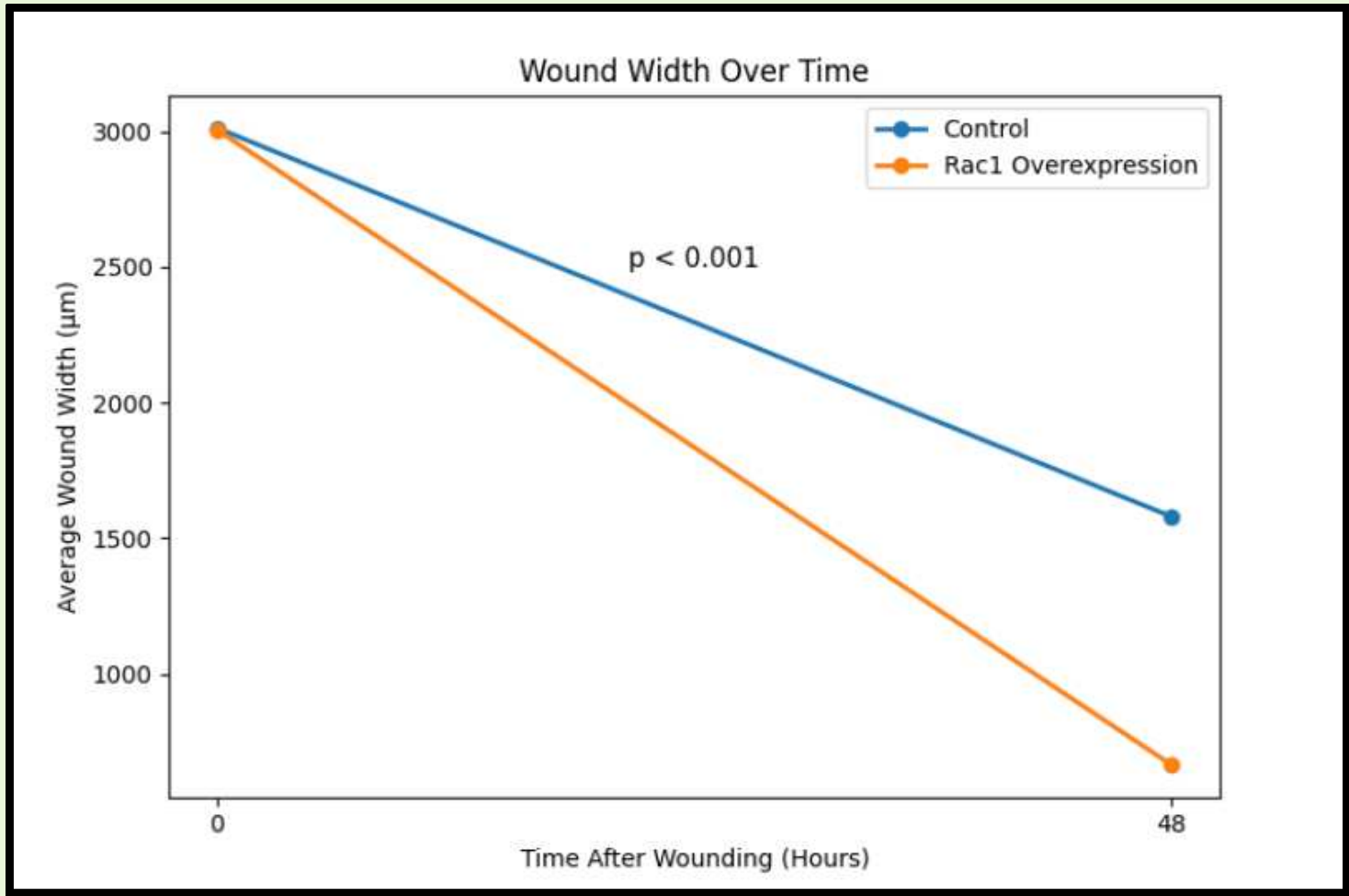
Data

Control Group — No Overexpression				
Sample	0 hr Wound Width (μm)	48 hr Wound Width (μm)	Change in Width (μm)	% Closure
1	3000	1980	1440	47.20%
2	2950	1920	1425	48.30%
3	3100	1950	1440	47.10%
4	2880	1480	1390	48.30%
5	3000	1960	1440	48.00%
6	3150	1700	1450	46.00%
7	2950	1910	1470	48.40%
8	3050	1600	1445	47.40%
9	2980	1930	1445	48.50%
10	3070	1620	1455	47.30%
11	2910	1500	1410	48.50%
12	3040	1980	1480	47.70%
13	2990	1950	1440	48.10%
14	3120	1960	1440	46.80%
15	2860	1470	1380	48.50%
16	3010	1910	1440	47.80%
17	3080	1630	1455	47.10%
18	3000	1920	1440	48.00%
19	3060	1610	1445	47.50%
20	3000	1980	1440	47.80%

Experimental Group — Rac1 Overexpression				
Sample	0 hr Wound Width (μm)	48 hr Wound Width (μm)	Change in Width (μm)	% Closure
1	3010	2010	1405	77.80%
2	2940	1910	1415	78.10%
3	2990	1700	1290	77.00%
4	2970	1860	1285	78.50%
5	3000	1890	1280	78.30%
6	3140	1920	1280	78.90%
7	2930	1820	1210	78.80%
8	3050	1700	1280	77.10%
9	2970	1820	1280	78.00%
10	2960	1780	1280	78.20%
11	2910	1800	1210	78.40%
12	3020	1870	1280	77.80%
13	2980	1860	1280	78.90%
14	3110	1780	1280	78.60%
15	2980	1860	1280	78.70%
16	3020	1860	1280	78.20%
17	3090	1730	1280	78.50%
18	2950	1810	1235	78.20%
19	3060	1720	1280	77.50%
20	2990	1840	1245	78.40%

Group	Average 0 hr Width (μm)	Average 48 hr Width (μm)	Average Reduction (μm)	Average % Closure
Control	3013	1979	1424	47.60%
Experimental	3006	1861	1245	78.00%

Results



An independent samples t-test was conducted to compare percent wound closure between the control group and the Rac1-overexpression group after 48 hours. The analysis revealed a statistically significant difference between the two groups ($t \sim -89.62$, $p < 0.001$), with the Rac1-overexpression group demonstrating substantially greater wound closure and smaller final wound widths than the control group.

Because the p-value was significantly lower than the accepted significance level of 0.05, the null hypothesis, which stated that there would be no significant difference in wound healing between the control and experimental groups, was rejected. These results support the alternative hypothesis that Rac1 overexpression enhances wound healing in *Drosophila melanogaster* larvae.

Overall, the findings suggest that Rac1 plays an important role in epithelial wound regeneration, likely by promoting actin cytoskeleton remodeling and coordinated cellular movement during tissue repair. The consistently faster healing observed in the experimental group indicates that increased Rac1 activity may accelerate wound closure. This project contributes to a broader understanding of the molecular mechanisms underlying tissue regeneration and demonstrates the value of *Drosophila melanogaster* as a model organism for studying wound healing processes.